

DEBABRATA SAHU

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B-302, HR-1, NIST University, Institute Park, Pallur Hills, Berhampur, Odisha, India - 761008



EDUCATION

NIST University

B.Tech - Electronics and Communication Engineering - CGPA: 7.7/10

2023 – 2027

Berhampur, India

Government Autonomous College

Intermediate - Science - Percentage: 70%

2020 – 2022

Rourkela, India

PROJECTS

Approximate Multiplier for AI Applications

08/2026 – Present

- Designed and implemented an approximate multiplier using Verilog to reduce hardware complexity while maintaining acceptable computational accuracy. Synthesized and evaluated the design using Xilinx Vivado with focus on hardware resources, timing, power, and approximation error.

Terahertz Metamaterial Absorber-Based Skin Cancer Detection System

05/2026 – 06/2026

- Designed and simulated a metamaterial absorber-based system for skin cancer detection using CST Studio Suite. Analyzed the electromagnetic response of the absorber to identify changes associated with abnormal skin tissue, with emphasis on sensing performance and detection characteristics.

Configurable Floating-Point FMA Unit using Mixed Precision

02/2026 – 04/2026

- Designed a configurable Floating-Point Fused Multiply-Add (FMA) unit using mixed precision for efficient floating-point arithmetic operations. Developed and simulated the RTL design using Verilog/SystemVerilog in Xilinx Vivado with emphasis on configurable precision and hardware-efficient computation.

Decentralized Networks

10/2025 – 11/2025

- Studied decentralized network architectures where control is distributed among multiple nodes to improve reliability, transparency, and fault tolerance. Analyzed working principles, security mechanisms, advantages, limitations, and real-world applications including blockchain, peer-to-peer systems, and decentralized finance.

12-Bit Analog-to-Digital Converter

08/2025 – 09/2025

- Designed and simulated a 12-bit Analog-to-Digital Converter (ADC) using Cadence Virtuoso for converting analog input signals into corresponding digital output signals. Performed schematic-level design and simulation to analyze the functionality and performance of the converter.

Smart Greenhouse Monitor System

05/2025 – 06/2025

- Developed a greenhouse monitoring system integrated with temperature, humidity, gas, and motion sensors for real-time environmental monitoring and automated control. Implemented servo motor-based automation with buzzer, LED, Wi-Fi, and Blynk for plant safety, remote monitoring, and efficient smart farming.

Smart Mining Helmet

03/2025 – 04/2025

- Developed a smart mining helmet integrated with heart rate, temperature, humidity, and gas sensors for real-time monitoring and worker safety. Implemented IoT connectivity using Wi-Fi, ThingSpeak, and Blynk for cloud data logging, remote monitoring, and automated alerts through buzzer, LED, and LCD.

INTERNSHIPS

Research Intern – NIT Jamshedpur <i>National Institute of Technology Jamshedpur</i>	18 May 2026 – 18 June 2026 <i>Jamshedpur, Jharkhand</i>
VLSI Project-Based Intern – EdXcellence <i>EdXcellence</i>	20 January 2026 – 18 March 2026
E&T Department Intern – Central Coalfields Limited <i>Central Coalfields Limited</i>	01 June 2025 – 30 June 2025 <i>Ranchi, Jharkhand</i>
MATLAB & Its Applications in Engineering <i>NIST University, Berhampur</i>	12 July 2024 – 1 August 2024 <i>Berhampur, Odisha</i>

TECHNICAL SKILLS

Languages: Verilog, SystemVerilog, Python, C, HTML, MATLAB, 8086 Assembly Language
Developer Tools: Cadence Virtuoso, Xilinx Vivado, VS Code, EDA Playground, Arduino IDE, ThingSpeak, MATLAB, TCAD, 8086 Assembly, Code Composer Studio, CST Studio Suite
Technologies/Frameworks: Digital VLSI Design, RTL Design, RTL Verification, FPGA, Embedded Systems, IoT, Microcontrollers (Arduino, ESP32), Microprocessor (8086), Sensor Interfacing, Hardware Debugging, Machine Learning

COURSES & TRAINING

FPGA-Based Communication System Design (FCSD-2026) <i>Electronics & ICT Academy, NIT Patna & NIT Jamshedpur</i>	25 May 2026 – 5 June 2026
– Faculty Development Program focused on FPGA-based communication system design and hardware implementation.	
Getting Started with Artificial Intelligence <i>IBM SkillsBuild</i>	Issued: 27 February 2026
– Completed foundational training covering fundamental concepts and applications of Artificial Intelligence.	
MATLAB and Its Applications in Engineering <i>NIST University, Berhampur</i>	12 July 2024 – 1 August 2024
– Completed practical training in MATLAB and its applications in engineering, including numerical analysis, simulation, and computational problem-solving.	

EXTRA-CURRICULAR ACTIVITIES

NBC – NIST Basketball Club <i>Player</i>	11 October 2023 – Present <i>NIST University, Berhampur</i>
Renewable Energy Club <i>Event Head • Social Media Head</i>	September 2024 – Present <i>NIST University, Berhampur</i>
Innovation & Incubation Club <i>Core Member</i>	November 2025 – Present <i>NIST University, Berhampur</i>